CLASS XII

PHYSICAL EDUCATION

EXTREMELY DETAILED NOTES

CBSE • Subject Code 048 • Academic Session 2026–27

Complete theory notes for all 10 units

Definitions • Explanations • Tables • Formulas • Tournament fixtures • Test procedures • Sports examples • Exam-focused recap

Prepared using the official CBSE Class XI–XII Physical Education curriculum for 2026–27 as the syllabus checklist. These are original study notes, not a replacement for teacher instructions or the prescribed textbook.

Theory: 70 marks | Practical/Internal Assessment: 30 marks | Total: 100 marks

How to use these notes

- Read the syllabus checklist at the start of each chapter and mark each topic after revising it.
- Learn definitions in precise language, then add one relevant sports example in long answers.
- For calculations and fixtures, practise drawing the table/diagram yourself; a worked pattern is included where useful.
- For asanas, tests, injuries and exercise, follow your teacher's demonstration and safe practice guidance. Do not attempt painful or medically unsuitable movements.

Syllabus overview

Unit	Chapter	Key areas
1	Management of Sporting Events	Functions, committees, fixtures, tournaments, community programmes
2	Children and Women in Sports	WHO activity guidance, posture, women in sport, menarche, menstrual dysfunction, female athlete triad
3	Yoga as a Preventive Measure for Lifestyle Diseases	Asanas for obesity, diabetes, asthma, hypertension, back pain and arthritis
4	Physical Education and Sports for CWSN	Disability and disorder, classification, inclusion, benefits, strategies, organisations
5	Sports and Nutrition	Balanced diet, nutrients, nutritive/non-nutritive components, weight management, sports diet
6	Test and Measurement in Sports	Fitness tests, BMI, BMR, senior citizen tests, motor educability
7	Physiology and Injuries in Sports	Exercise effects, ageing, injury types, prevention and management
8	Biomechanics and Sports	Newton's laws, levers, equilibrium, friction, projectile motion
9	Psychology and Sports	Personality, motivation, aggression, exercise adherence, psychological attributes
10	Training in Sports	Talent identification, training cycles, strength, endurance, speed, flexibility, coordination, circuit training

Exam point: The official syllabus has 10 theory units. The precise unit-wise marks pattern can vary by the published blueprint; use the latest school/CBSE exam instructions for final weightage.

CHAPTER 1 | Management of Sporting Events

This unit focuses on planning and conducting sports events efficiently, fairly and safely.

1.1 Functions of sports-event management

Function	Meaning	Sports-event example
Planning	Deciding objectives, date, venue, budget, schedule, equipment, officials and safety arrangements before the event.	Prepare the timetable, budget and venue plan for an annual sports day.
Organising	Arranging resources and assigning work so the plan can be carried out.	Arrange teams, equipment, playing areas and event sequence.
Staffing	Selecting, recruiting, briefing and assigning people with suitable skills.	Appoint referees, scorers, volunteers, first-aid staff and announcers.
Directing	Guiding, instructing, motivating and communicating with workers during preparation and the event.	Give volunteers clear instructions and solve immediate coordination issues.
Controlling	Comparing actual progress with the plan, monitoring standards and correcting deviations.	Check time, expenses, safety, rule compliance and event progress.

1.2 Sports committees and responsibilities

Committees divide work among people with specific duties. The exact committee structure depends on the size and nature of the event.

Committee	Typical responsibilities
Organising / executive	Overall coordination, approvals, decisions and supervision.
Finance	Budget, accounts, payments, receipts, sponsorship records and final statement.
Ground and equipment	Prepare fields/courts, mark boundaries, arrange and inspect equipment.
Technical	Rules, fixtures, officials, eligibility, results and technical disputes.
Boarding and lodging	Accommodation, meals, water and hospitality for participants/officials.
Transport	Travel plans, pick-up/drop-off schedules, vehicle coordination.
Reception	Welcome guests, teams and officials; provide information and guidance.
Medical / first aid	First-aid station, emergency contact plan, medical support and referral arrangements.
Publicity	Notices, invitations, posters, announcements, media and result updates.
Refreshment	Safe drinking water and refreshments at appropriate times.
Discipline and security	Crowd management, access control, participant conduct and emergency procedures.
Prize / ceremony	Certificates, medals, trophies, opening and closing ceremonies.
Documentation	Attendance, permissions, result sheets, photographs, reports and records.

1.3 Committees before, during and after the event

- **Before:** select venue/date, make budget, obtain permissions, prepare fixtures, appoint officials, inspect facilities, publicise the event and plan safety.
- **During:** coordinate events, keep time, record scores, manage spectators, provide water/medical support, resolve rule-related issues and communicate results.
- **After:** announce verified results, distribute awards, return equipment, settle accounts, clean the venue, thank contributors and prepare an evaluation report.

1.4 Tournament fixtures

A **fixture** is the schedule or arrangement showing which teams/players compete, and when or in which round. A tournament format determines how a winner is decided.

A. Knock-out tournament

- A team is eliminated after losing one match; winners advance to the next round.
- Advantages: fewer matches, lower cost, shorter duration, useful when many teams enter.
- Limitations: a strong team may be eliminated after one poor performance; teams get unequal match opportunities.

Formula: Number of matches = N – 1, where N is the number of teams.

Byes: A bye allows a team to move to the next round without playing in the first round. Number of byes = next power of 2 – N. If N is already a power of 2, there are no byes.

Example: 11 teams; next power of 2 is 16; byes = 16 – 11 = 5; matches = 11 – 1 = 10.

Drawing procedure: determine the number of teams; find the next power of 2; calculate byes; distribute byes fairly (usually between upper/lower halves and quarters); place seeded teams in separate positions; fill remaining teams by lot; draw the bracket.

B. Seeding

Seeding means placing the strongest or highest-ranked teams in positions intended to prevent them from meeting in the earliest rounds. It improves competitive fairness when reliable rankings are available. It does not guarantee a team will win.

C. League / round-robin tournament

- Each team plays every other team. It gives teams more matches and usually a fairer indication of overall performance.
- Limitations: requires more time, facilities and expenditure.

Single league formula: Matches = N(N – 1) ÷ 2. **Double league:** Matches = N(N – 1).

For 6 teams, single league matches = 6 × 5 ÷ 2 = 15.

League method	How it works
Staircase method	List teams in a stair-step pattern; each team is paired with every team appearing below it. Easy for small numbers of teams.
Cyclic method	Keep one team fixed and rotate the others each round. Used to generate round-robin rounds.
Tabular method	Use a table/matrix with teams on rows and columns; fill each pair's match once. Helpful for checking that no pairing is missed.

D. Combination tournament

A combination tournament uses two or more formats, commonly league within groups followed by knock-out rounds. It balances participation and time. Example: 16 teams are divided into four groups; each group plays a league; the top team(s) advance to a knock-out stage.

1.5 Intramural and extramural tournaments

Basis	Intramural	Extramural
Meaning	Competitions within one institution/school.	Competitions between different institutions/schools/clubs.
Main aim	Mass participation, recreation, fitness, leadership and house spirit.	Higher-level competition, exposure, performance comparison and inter-school relations.
Participation	Usually open to eligible members of the same institution.	Selected teams/representatives compete against outside institutions.
Examples	Inter-house football or athletics day.	Inter-school basketball tournament.

1.6 Community sports programmes

- **Sports Day:** planned programme of races, games and demonstrations; encourages participation, fitness, teamwork and community spirit.
- **Health Run:** a non-elite mass run that promotes awareness of health and physical activity.
- **Run for Fun:** participation-focused run intended for enjoyment, social connection and active living rather than winning.
- **Run for a Specific Cause:** event organised to raise awareness or support a cause, such as environmental protection or public health.
- **Run for Unity:** community run that promotes solidarity, inclusion and social harmony.

Exam point: Long-answer structure: define the concept → list/describe the points → give a practical example → mention importance or limitations.

CHAPTER 2 | Children and Women in Sports

2.1 WHO physical-activity guidance: key age groups

The curriculum expects students to know exercise guidance for different age groups. WHO recommendations are population-level guidance, not a competition prescription.

Age group	General guidance to remember
Children and adolescents (5–17 years)	Average at least 60 minutes per day of moderate-to-vigorous physical activity across the week, mostly aerobic; include vigorous aerobic and muscle- and bone-strengthening activities on at least 3 days/week.
Adults (18–64 years)	150–300 minutes of moderate-intensity aerobic activity per week, or 75–150 minutes vigorous activity, or an equivalent combination; muscle-strengthening activities on 2 or more days/week.
Older adults (65+ years)	Adult aerobic and strengthening guidance, adapted to ability; include varied multicomponent activity emphasising functional balance and strength on 3 or more days/week to support capacity and prevent falls.

Activity should be age-appropriate, enjoyable, accessible and increased gradually. Anyone with a health concern should seek appropriate professional advice before changing exercise substantially.

2.2 Common postural deformities and corrective measures

Posture is the alignment of body parts while standing, sitting or moving. A postural deformity is an abnormal alignment that may be structural or functional. Exercises may help some functional problems, but structural deformities require professional assessment.

Deformity	Meaning / appearance	General corrective or management measures
Knock knees (genu valgum)	Knees angle inward and may touch while ankles remain apart.	Professional assessment if persistent/painful; prescribed strengthening and movement training; maintain suitable footwear and avoid self-prescribed corrective devices.
Flat foot	Arch of the foot is low or absent when standing.	Foot-muscle strengthening and calf stretching if advised; supportive footwear/orthoses when recommended; assess pain or gait difficulty.
Round shoulders	Shoulders roll forward, often with forward head posture.	Posture awareness; strengthening of upper-back muscles; mobility work for chest/shoulder muscles; ergonomic sitting and screen habits.
Lordosis	Excessive inward curve of the lumbar spine.	Assessment of cause; trunk and hip strengthening/mobility programme prescribed to the individual; avoid assuming all spinal curves are correctable with one exercise.
Kyphosis	Excessive outward curve of the upper back.	Postural training and back-extensor strengthening when appropriate; medical assessment for pain, marked curve or suspected structural cause.
Scoliosis	Sideways spinal curve, sometimes with rotation.	Screening/medical assessment; exercises or bracing may be recommended depending on severity and growth; do not attempt forceful self-correction.
Bow legs (genu varum)	Legs curve outward; ankles may meet while knees remain apart.	Assessment for age, cause and severity; appropriate strengthening and clinical management; persistent or painful cases need professional review.

2.3 Benefits of women’s participation in sports

Physical benefits

- Improves cardiovascular endurance, muscular strength, flexibility, coordination and bone health.
- Supports healthy blood pressure, glucose regulation, sleep and long-term physical function.
- Develops motor skills, balance, agility and confidence in movement.

Psychological benefits

- Can improve self-confidence, self-efficacy, mood, stress management and body competence.
- Develops goal setting, persistence, decision-making and emotional regulation.
- Provides experiences of achievement and healthy coping with success and failure.

Social benefits

- Encourages teamwork, leadership, communication, friendship and belonging.
- Challenges stereotypes and expands opportunities for education, careers and public participation.
- Builds visibility of women as athletes, coaches, officials, administrators and role models.

2.4 Menarche and menstrual dysfunction

Menarche is the first menstrual period, marking a stage of reproductive maturation. Timing varies between individuals. Menstrual cycles may be irregular in the first years after menarche, but persistent irregularity, very heavy bleeding, severe pain or missed periods should be discussed with a qualified healthcare professional.

Menstrual dysfunction is a broad term for disturbances in menstrual timing, frequency, duration or symptoms. Examples include amenorrhea (absence of menstruation), dysmenorrhea (painful menstruation) and unusually heavy bleeding. In athletes, low energy availability, high training load, stress and some medical conditions may contribute.

- Supportive sport practice includes privacy, access to menstrual products and toilets, flexible support when needed, appropriate nutrition and recovery, and non-stigmatising communication.

2.5 Female Athlete Triad

The traditional Female Athlete Triad describes three interrelated conditions:

Component	Meaning	Why it matters
Low energy availability / disordered eating	Energy intake is insufficient relative to exercise needs; it may occur unintentionally or alongside disordered eating.	Can impair recovery, health, concentration and performance.
Menstrual dysfunction / amenorrhea	Menstrual cycles become irregular or periods stop.	May indicate that the body's energy and hormonal systems are under strain.
Low bone mineral density / osteoporosis	Bones may become less dense and more vulnerable to stress injuries or fractures.	Can affect present health and future bone strength.

The components can reinforce one another. Prevention includes adequate energy and nutrient intake, balanced training and rest, early recognition of symptoms, and support from healthcare and sports professionals. This is a health topic—not a reason to judge an athlete's body or encourage restrictive eating.

Exam point: Remember: Female Athlete Triad = energy/eating problem + menstrual dysfunction + impaired bone health.

CHAPTER 3 | Yoga as a Preventive Measure for Lifestyle Diseases

3.1 Meaning and role of yoga

Yoga is a mind–body discipline involving physical postures (asanas), breathing practices (pranayama), relaxation and attention. It can complement healthy eating, physical activity, sleep and medical care. Yoga is not a guaranteed cure and must not replace prescribed treatment.

3.2 Lifestyle diseases in the syllabus

Lifestyle diseases are conditions whose risk is influenced by behaviours and environment, including inactivity, diet, stress and sleep. The syllabus focuses on obesity, diabetes, asthma, hypertension, back pain and arthritis.

3.3 Asanas commonly prescribed in Class XII

The asanas below are commonly taught in school Physical Education resources for these conditions. Different textbooks may pair slightly different asanas with a condition; follow your school's prescribed list. Practice under qualified supervision, avoid pain and observe contraindications.

Condition	Commonly listed asanas / practices	Potential supportive role and caution
Obesity	Tadasana, Katichakrasana, Pavanamuktasana, Bhujangasana, Dhanurasana, Paschimottanasana (lists may vary).	May support mobility, activity habits and stress management; weight management also requires sustainable nutrition and activity habits.
Diabetes	Mandukasana, Ardha Matsyendrasana, Vakrasana, Bhujangasana, Dhanurasana, Kapalabhati where taught.	Yoga may support stress management and general activity. People using glucose-lowering medicine should consider hypoglycaemia risk and follow medical advice.
Asthma	Sukhasana, Matsyasana, Bhujangasana, Gomukhasana, Anulom-Vilom or other gentle breathing practice.	Gentle breathing/relaxation may support symptom coping. Do not force breath retention; follow an asthma action plan and never replace inhalers with yoga.
Hypertension	Shavasana, Makarasana, Sukhasana, Vajrasana and gentle relaxation/breathing.	Relaxation may help stress management. Avoid forceful breathing, straining and inversions unless a qualified clinician/teacher says appropriate.
Back pain	Makarasana, Bhujangasana, Shalabhasana, Ardha Chakrasana or other gentle mobility exercises as prescribed.	Selection depends on the cause of pain. Avoid any posture that worsens symptoms; severe, persistent or radiating pain needs assessment.
Arthritis	Tadasana, gentle joint-mobility movements, Vajrasana only if comfortable, and supported relaxation.	Low-impact movement can maintain mobility, but joint-specific modifications are important. Avoid loading an inflamed or painful joint.

3.4 Important asana notes

Tadasana (Mountain Pose)

Stand tall with feet stable, spine long and shoulders relaxed; distribute weight evenly. Avoid locking the knees.

Katichakrasana (Standing spinal twist)

Stand with feet apart and rotate the trunk gently within a comfortable range; do not jerk or force the neck.

Bhujangasana (Cobra Pose)

Lie prone, place hands near the chest and lift the upper trunk gently using back muscles; do not push into lower-back pain.

Pavanamuktasana (Wind-relieving Pose)

Lie on the back and bring a bent knee toward the torso only as comfortable; avoid forceful compression or if contraindicated.

Mandukasana (Frog Pose variation)

A kneeling forward-flexion posture taught in several variations; follow the teacher's exact version and avoid if knees or hips are painful.

Ardha Matsyendrasana (Half spinal twist)

A seated twist performed gently with an upright spine; avoid forcing rotation, especially with spinal problems.

Anulom-Vilom

Alternate-nostril breathing at a comfortable pace; do not hold the breath forcefully.

Shavasana

Lie comfortably, release unnecessary muscular tension and observe natural breathing; stop if lying flat causes discomfort.

Makarasana

Prone relaxation posture with the head and arms supported comfortably; avoid pressure that aggravates breathing or back symptoms.

3.5 General precautions

- Learn correct alignment and breathing from a qualified instructor.
- Avoid yoga immediately after a heavy meal; use a comfortable, ventilated space.
- Stop if there is dizziness, chest discomfort, unusual breathlessness or sharp pain.
- People with pregnancy, recent surgery, uncontrolled blood pressure, significant spinal/joint conditions or other health concerns need individual advice.

Exam point: For a 5-mark answer, write the disease, name 4–6 relevant practices from the school list, explain their supportive role and state one safety precaution.

CHAPTER 4 | Physical Education and Sports for CWSN

4.1 Key concepts

CWSN means Children With Special Needs. The term covers children who may need adaptations or additional support to participate fully in learning and physical activity.

Term	Meaning
Disability	A long-term physical, sensory, intellectual or psychosocial impairment that, in interaction with barriers, may restrict full participation.
Disorder	A health or developmental condition that may affect functioning; it is not synonymous with every disability.
Impairment	A difference or loss in body structure or function.
Activity limitation	Difficulty carrying out a task or action.
Participation restriction	Difficulty being involved in a life situation because of personal and environmental factors.
Adaptive Physical Education	An individualised programme that modifies activities, equipment, instruction or environment to meet learners' abilities and needs.

4.2 Types of disability and examples

- **Physical / mobility disability:** may affect movement, balance, coordination or endurance.
- **Visual impairment:** reduced vision or blindness; sound cues, tactile markings and clear verbal descriptions may help.
- **Hearing impairment:** reduced hearing; visual signals, demonstrations and written instructions may help.
- **Intellectual disability:** limitations in intellectual functioning and adaptive behaviour; instruction may need to be concrete, repeated and broken into steps.
- **Autism spectrum and other neurodevelopmental needs:** individual differences in communication, sensory processing, routines and social interaction may call for predictable, flexible instruction.

Causes may be congenital, genetic, developmental, illness-related or acquired through injury. Never assume that all people with the same diagnosis have the same needs or abilities.

4.3 Disability etiquette

- Speak directly to the person, not only to a companion or caregiver.
- Ask before offering help; do not touch a wheelchair, cane or other mobility aid without permission.
- Use respectful language and avoid pity, teasing, stereotypes or unwanted personal questions.
- Give enough time for responses and explain rules or changes clearly.
- For a visually impaired participant, identify yourself, describe the space and ask how guidance should be offered.
- For a deaf/hard-of-hearing participant, gain attention respectfully and use agreed visual or written signals.

4.4 Aims and objectives of Adaptive Physical Education

- Provide safe, meaningful and enjoyable opportunities for physical activity.
- Develop physical fitness, fundamental movement skills, coordination and functional independence.
- Promote social interaction, confidence, self-expression, leadership and emotional wellbeing.
- Adapt tasks to individual strengths and needs rather than excluding a learner because of a disability.
- Support lifelong participation in recreation, sport and healthy living.

4.5 Benefits of physical activity for CWSN

Area	Benefits
Physical	Supports strength, endurance, mobility, balance, coordination and functional skills.
Psychological	Can build confidence, self-efficacy, enjoyment, stress regulation and a sense of achievement.
Social	Creates opportunities for friendships, teamwork, communication and belonging.
Cognitive / learning	Can support attention, task-following, problem-solving and learning through movement.
Health	Contributes to active living and helps reduce risks associated with prolonged inactivity.

4.6 Inclusion in sports

Inclusion means designing environments where people with diverse abilities can participate meaningfully together. It is more than simply allowing a participant to attend.

- Modify **rules** (e.g., extra time, simplified scoring or adapted boundaries where appropriate).
- Modify **equipment** (lighter/larger balls, audible balls, contrast markings, suitable grips).
- Modify **environment** (ramps, clear routes, accessible toilets, safe surfaces, quiet space).
- Modify **instruction** (demonstration, verbal cues, pictograms, one-step instructions, repetition).
- Use peer buddies when welcomed by the participant, while preserving independence and choice.
- Assess individual needs and risks; adaptations should preserve the purpose of the activity and dignity of participants.

4.7 Classification and divisioning

Classification in disability sport groups athletes according to the impact of eligible impairments on the activities required by a sport, aiming for meaningful competition. Rules differ by sport and governing body.

Divisioning, especially associated with Special Olympics, groups athletes for competition using factors such as ability/performance, age and sex as required by event rules, so competition is as equitable as possible.

4.8 Organisations and sports

- **Paralympic movement:** international sport for eligible athletes with disabilities, governed by the International Paralympic Committee and sport-specific bodies.
- **Special Olympics:** provides year-round sports training and competition opportunities for people with intellectual disabilities.
- **Deaf sport:** international competition for deaf athletes, including the Deaflympics.
- Examples of adapted/inclusive sports: goalball, boccia/bocce, wheelchair basketball, sitting volleyball, blind cricket and unified sports.

Exam point: High-scoring answers mention the barrier, the adaptation and the participation benefit—not just the disability label.

CHAPTER 5 | Sports and Nutrition

5.1 Meaning and importance of nutrition

Nutrition is the process by which the body obtains and uses nutrients from food. A balanced diet provides energy and essential nutrients in suitable amounts for growth, repair, health and activity.

5.2 Balanced diet

A balanced diet includes carbohydrates, proteins, fats, vitamins, minerals, water and fibre in appropriate amounts, adjusted for age, growth, health, culture, activity and access to food.

- Energy needs vary with body size, growth, activity level and health; there is no single diet suitable for every athlete.
- Variety across food groups helps reduce the risk of nutrient gaps.
- Regular meals, hydration and adequate recovery support learning and sports participation.

5.3 Macronutrients

Nutrient	Main functions	Sources / notes
Carbohydrates	Major energy source, especially for moderate-to-high intensity activity; stored as glycogen in muscles and liver.	Grains, rice, roti, oats, potatoes, fruit, pulses and dairy. Fibre-rich sources support gut health.
Proteins	Growth and repair of tissues; enzymes, hormones and immune functions; muscle repair after training.	Pulses, beans, lentils, soy, dairy, eggs, fish, meat, nuts and seeds. Needs depend on age, training and total diet.
Fats	Concentrated energy, cell membranes, hormone-related functions and absorption of vitamins A, D, E and K.	Nuts, seeds, oils, avocado and other foods. Include unsaturated fats; avoid treating all fats as harmful.

5.4 Micronutrients

Nutrient	Functions	Food examples / deficiency context
Vitamin A	Vision, immune function and epithelial health.	Carrots, leafy vegetables, dairy, eggs; deficiency can affect vision.
B-group vitamins	Energy metabolism, nerve function and red blood cell formation (roles vary by vitamin).	Whole grains, pulses, dairy, eggs, meat and leafy vegetables; deficiency depends on the vitamin.
Vitamin C	Collagen formation, wound healing and antioxidant function; helps iron absorption.	Citrus, guava, amla, tomatoes and peppers.
Vitamin D	Calcium absorption and bone health.	Sunlight exposure and selected foods/fortified foods; deficiency can affect bones.
Vitamin E	Antioxidant role in cell membranes.	Nuts, seeds and vegetable oils.
Vitamin K	Normal blood clotting and bone-related functions.	Green leafy vegetables and other foods.
Calcium	Bone and teeth structure; muscle contraction and nerve signalling.	Dairy, calcium-set tofu, selected greens and fortified foods.
Iron	Haemoglobin formation and oxygen transport.	Pulses, meat, leafy vegetables and fortified foods; low iron may cause fatigue.
Iodine	Thyroid hormone production.	Iodised salt and selected foods.
Sodium / potassium	Fluid balance, nerve signalling and muscle function.	Varied foods; losses in sweat differ with conditions. Avoid unnecessary salt loading.

5.5 Water and fibre

- **Water:** supports temperature regulation, circulation, digestion and transport of nutrients. Fluid needs increase with heat, duration and sweat loss.

- **Dietary fibre:** supports bowel regularity and gut health; found in whole grains, pulses, vegetables, fruit, nuts and seeds.

5.6 Nutritive and non-nutritive components

Nutritive components provide energy and/or essential substances used by the body: carbohydrates, proteins, fats, vitamins, minerals and water. **Non-nutritive components** do not supply essential nutrients in the same way but may affect food quality or body function, e.g., fibre, pigments, flavour compounds and some plant compounds. Classification can vary by textbook; fibre is often described as a non-nutritive food component despite its important health benefits.

5.7 Food myths and weight control

- Body weight is affected by genetics, growth, sleep, medication, health, environment, food access and activity—not willpower alone.
- Crash diets, skipping meals, dehydration and unprescribed supplements can harm health and performance.
- Healthy weight management is gradual and individualised, and focuses on energy, strength, wellbeing and sustainable habits rather than appearance.
- For teenagers, growth and development make restrictive dieting especially risky; consult a qualified healthcare professional for individual concerns.

5.8 Nutrition before, during and after exercise

Timing	General principles
Before activity	Eat familiar, easily tolerated food with carbohydrate; allow enough time for digestion. Avoid experimenting with unfamiliar foods just before competition.
During activity	For short sessions, water is usually sufficient. Longer, vigorous events or hot conditions may require planned fluids and carbohydrate, guided by duration and individual needs.
After activity	Rehydrate, eat a normal meal/snack containing carbohydrate and protein, and continue regular meals to support recovery.
Everyday athlete diet	Prioritise adequate total energy, variety, carbohydrate for training, protein across meals, micronutrients, hydration and sufficient sleep.

5.9 Common nutrition-related terms

- **Malnutrition:** poor nutrition from insufficient, excessive or imbalanced intake; it includes undernutrition and overnutrition.
- **Dehydration:** excessive loss of body water, which can impair wellbeing and performance.
- **Ergogenic aid:** a method or product claimed to enhance performance; not all are effective or safe, and supplements may be contaminated.

Exam point: For exam answers, pair each nutrient with function and food source. Do not confuse energy-giving macronutrients with vitamins and minerals.

CHAPTER 6 | Test and Measurement in Sports

6.1 Test, measurement and evaluation

Term	Definition	Example
Test	A procedure or task used to assess a particular ability or attribute.	A 50 m sprint test.
Measurement	Quantifying the result using a score, time, distance or other unit.	Sprint time = 8.4 seconds.
Evaluation	Interpreting measurements against standards or goals to make a judgement or decision.	Compare the time with an appropriate age/sex reference or the athlete's prior result.

6.2 Importance

- Identifies strengths and areas for improvement; helps set goals and design training.
- Tracks progress and evaluates the effectiveness of a programme.
- Supports selection, classification and safe monitoring when tests are valid and fairly administered.
- Encourages objective feedback and comparison with suitable standards.

6.3 SAI Khelo India Fitness Assessment (school-level overview)

The commonly taught age-appropriate school battery assesses health-related and skill-related fitness. Exact age bands, procedures and scoring tables should be checked against the current CBSE/SAI manual used by the school.

Component / test	What it assesses	Key administration idea
BMI	Body mass relative to height.	Measure height and weight accurately; calculate BMI. For children/adolescents, interpret with age- and sex-specific references rather than adult cut-offs.
50 m sprint	Speed.	Start from a standard line, sprint through finish; record time consistently.
600 m run/walk	Cardiorespiratory endurance.	Complete the distance at a sustainable effort; record time and follow the standard protocol.
Sit-and-reach	Flexibility, particularly hamstring/lower-back flexibility.	Use the prescribed box/setup; warm up and avoid bouncing.
Partial curl-ups	Abdominal muscular strength/endurance.	Use standard cadence, range and stopping rules.
Push-ups / modified push-ups	Upper-body muscular strength/endurance.	Use the prescribed form and age-appropriate version.
Standing broad jump	Explosive leg power.	Jump forward from behind the line; measure the best valid distance.

6.4 Harvard Step Test

The Harvard Step Test estimates cardiovascular fitness through recovery heart rate after stepping at a standard height and cadence. School protocols may vary by version, so use the exact test duration, step height and cadence taught by the teacher.

Common fitness index formula: Fitness Index = (exercise duration in seconds × 100) ÷ [2 × sum of recovery pulse counts]. The recovery pulse counts and time windows must match the version of the protocol being used.

Interpretation: a higher fitness index generally indicates better recovery in this test, but comparisons are meaningful only when the same standard protocol is used.

6.5 Basal Metabolic Rate (BMR)

BMR is the energy used by the body at rest to maintain vital functions under standard resting conditions. It is influenced by body size, age, sex, body composition, hormones, health and other factors.

In many school texts, a simplified estimation formula is taught. If your teacher or prescribed textbook specifies a formula, use that exact formula in examinations; formula variants exist.

Important distinction: BMR is not the same as total daily energy expenditure, which also includes movement, digestion and other activity.

6.6 Rikli and Jones Senior Citizen Fitness Test

This test battery evaluates functional fitness in older adults, using practical tasks relevant to daily living.

Test	Ability assessed	Basic procedure concept
Chair Stand Test (30 seconds)	Lower-body strength.	Count full stands from a chair in 30 seconds using standard technique.
Arm Curl Test (30 seconds)	Upper-body strength.	Count controlled curls with the prescribed weight in 30 seconds.
Chair Sit-and-Reach	Lower-body flexibility.	From seated position, reach toward the extended leg/foot and measure distance.
Back Scratch Test	Upper-body/shoulder flexibility.	Reach one hand over shoulder and the other behind the back; measure the gap/overlap.
8-Foot Up-and-Go	Agility and dynamic balance.	Rise from chair, walk around marker and return; record time.
6-Minute Walk Test or 2-Minute Step Test	Aerobic endurance.	Measure distance walked in six minutes or count steps in two minutes according to protocol.

6.7 Johnson–Metheny Test of Motor Educability

Motor educability refers to the ability to learn new motor skills. The Johnson–Metheny test is a battery of movement tasks used to assess how readily a learner acquires and performs unfamiliar movement patterns. Follow the prescribed manual for the exact items, demonstrations, scoring and sequence; do not invent a scoring rule when the school's version differs.

6.8 BMI and BMR: concepts and formulas

BMI formula: BMI = weight in kilograms ÷ (height in metres)^2.

Illustration: weight 60 kg, height 1.70 m $\rightarrow$ BMI = 60 ÷ (1.70 × 1.70) $\approx$ 20.8 kg/m^2. This is a mathematical illustration, not a diagnosis. For adolescents, BMI interpretation requires age- and sex-specific growth charts.

Reliability: consistency of results across repeated measurements. **Validity:** whether the test measures what it claims to measure. **Objectivity:** results are minimally affected by the tester's personal judgement.

Exam point: Learn: test = tool/task; measurement = number; evaluation = meaning/judgement. Use standardised procedures for valid comparisons.

CHAPTER 7 | Physiology and Injuries in Sports

7.1 Meaning

Physiology studies how the body functions. Exercise physiology examines acute responses and long-term adaptations to physical activity and training.

7.2 Physiological factors determining components of physical fitness

Fitness component	Important physiological influences
Strength	Muscle size and architecture, nervous-system recruitment, muscle fibre characteristics, technique, age and training history.
Speed	Nervous-system reaction and coordination, force production, fast muscle fibres, technique and energy availability.
Endurance	Heart and lung function, oxygen transport, capillary supply, mitochondrial capacity, fuel availability and pacing.
Flexibility	Joint structure, muscle-tendon properties, temperature, age and the nervous system's tolerance of stretch.
Coordinative ability	Sensory information, nervous-system processing, timing, balance, practice and movement experience.

7.3 Effects of exercise on the muscular system

- Regular strength training can improve muscle strength, power and, with suitable programming, muscle size.
- Repeated activity improves neuromuscular coordination and the ability to recruit muscles efficiently.
- Tendons and connective tissues may adapt gradually; loading should progress rather than jump suddenly.
- Exercise increases local blood flow and energy demand; fatigue develops when demand exceeds the body's ability to sustain output.

7.4 Effects on the cardiovascular and respiratory systems

System	Acute response during exercise	Long-term adaptation with regular training
Cardiovascular	Heart rate, stroke volume and cardiac output rise; blood flow is redistributed toward active muscles.	At a given submaximal workload, heart rate may be lower; stroke volume and aerobic capacity may improve.
Respiratory	Breathing rate and depth increase to supply oxygen and remove carbon dioxide.	Ventilatory efficiency and ability to sustain exercise may improve; responses depend on training and health.
Blood / circulation	Oxygen delivery and heat transport increase; sweating supports cooling.	Aerobic training can improve oxygen transport capacity and peripheral use, depending on the programme.

Cardiac output = heart rate × stroke volume. It is the volume of blood pumped by the heart per minute.

7.5 Physiological changes due to ageing

- Maximum aerobic capacity, muscle mass and strength may decline with age, especially without regular activity.
- Balance, reaction time, joint mobility and recovery may change; the rate varies greatly among individuals.
- Regular, appropriately designed aerobic, strength, balance and mobility activity supports independence and health at older ages.
- Training should be progressive and adapted to health, medication, previous activity and personal goals.

7.6 Sports injuries: classification

Type	Description / example
Soft-tissue injury	Affects muscles, tendons, ligaments, skin or other soft tissues; e.g., strain, sprain, bruise.
Hard-tissue injury	Affects bones or joints; e.g., fracture or dislocation.
Sprain	Stretching or tearing of a ligament around a joint.
Strain	Overstretching or tearing of muscle or tendon.
Abrasion	Surface scrape from friction.
Contusion	Bruise caused by impact.
Fracture	Break or crack in a bone.
Dislocation	A bone is displaced from its normal position at a joint; needs urgent professional care.

7.7 Causes and prevention of injuries

- Causes: poor technique, unsuitable equipment or surfaces, inadequate preparation, fatigue, sudden workload increase, collisions and ignoring pain.
- Prevention: appropriate warm-up, progressive training, technique instruction, suitable footwear/equipment, hydration, rest, safe surfaces and adherence to rules.

7.8 First response and PRICE

Some school resources use **PRICE** for initial management of certain soft-tissue injuries: **Protection, Rest, Ice, Compression, Elevation.** This is not a substitute for clinical assessment, and it is not appropriate for every injury.

- Protect the area and stop the activity; do not force movement or try to put a dislocated joint back.
- Seek urgent medical help for suspected fracture/dislocation, severe pain or swelling, deformity, inability to bear weight, head/neck injury, breathing difficulty or loss of consciousness.
- Ice should be wrapped rather than placed directly on skin. Compression must not be tight enough to cause numbness, colour change or increased pain.
- Return to sport should be gradual and based on recovery and appropriate professional guidance.

Exam point: Differentiate sprain (ligament) from strain (muscle/tendon)—a common short-answer question.

CHAPTER 8 | Biomechanics and Sports

8.1 Meaning and importance

Biomechanics applies mechanical principles to living bodies and movement. In sport, it helps explain technique, performance, equipment design and injury reduction. **Kinesiology** is the study of human movement, drawing on anatomy, physiology and mechanics.

8.2 Newton’s laws of motion

Law	Explanation	Sports example
First law: inertia	An object remains at rest or continues in uniform straight-line motion unless acted on by a net external force.	A football stays still until kicked; a moving skater tends to continue moving.
Second law: acceleration	Acceleration depends on net force and mass; F = m × a.	A stronger net force can accelerate a ball more; for the same force, a lighter object accelerates more.
Third law: action–reaction	For every action force, there is an equal and opposite reaction force acting on another body.	A runner pushes backward on the ground; the ground exerts a forward force on the runner.

8.3 Levers in the human body

A lever is a rigid bar that rotates around a fulcrum. In the body, bones act as levers, joints as fulcrums and muscles provide force.

Class	Arrangement	Example
First-class	Fulcrum lies between effort and resistance.	Nodding the head at the atlanto-occipital joint; elbow extension by triceps is often taught as an example.
Second-class	Resistance lies between fulcrum and effort.	Rising onto the toes: ball of foot = fulcrum, body weight = resistance, calf muscles = effort.
Third-class	Effort lies between fulcrum and resistance.	Biceps flexing the elbow while lifting a load; common in the human body and favours speed/range of motion.

8.4 Equilibrium and centre of gravity

Equilibrium is a state in which the net force and net torque on a body are zero. **Centre of gravity** is the point through which the body’s weight may be considered to act.

- Static equilibrium: body remains at rest (e.g., a held balance pose).
- Dynamic equilibrium: body remains balanced while moving or changing position.
- Stability generally increases with a wider base of support, a lower centre of gravity and a line of gravity that remains within the base of support.
- A player may lower the body and widen the stance to resist an opponent’s push.

8.5 Friction in sports

Friction is a force that opposes relative motion or the tendency to move between surfaces.

- **Helpful friction:** shoe-ground grip for sprinting, hand grip on equipment, traction for changing direction.
- **Unhelpful friction:** air resistance that slows a fast-moving object or athlete; friction can also cause skin abrasions or heat.
- Friction depends on surfaces, normal force and other conditions. Cleats, shoe tread, surface materials and equipment textures can alter grip.

8.6 Projectile motion

Projectile motion occurs when an object is launched into the air and then moves mainly under gravity, with air resistance often simplified or ignored in basic models.

- The horizontal and vertical components of motion are analysed separately.
- For a given launch speed on level ground in an ideal model, range depends on launch angle; 45° gives maximum range only under simplified assumptions of equal launch and landing height and no air resistance.
- Real sporting performance also depends on release height, air resistance, spin, equipment and the athlete's technique.

Examples: javelin throw, long jump take-off, basketball shot and a kicked ball.

8.7 Common movement terms

Term	Meaning	Example
Flexion	Decreases the angle at a joint.	Bending the elbow.
Extension	Increases the angle at a joint / straightens a joint.	Straightening the knee.
Abduction	Moves a body part away from the midline.	Lifting the arm sideways.
Adduction	Moves a body part toward the midline.	Bringing the arm back to the side.
Rotation	Turns around a longitudinal axis.	Turning the head left/right.
Circumduction	Circular movement combining flexion, extension, abduction and adduction.	Arm circles at the shoulder.
Supination	Forearm rotates so palm faces upward/anteriorly in anatomical position.	Holding a bowl in the palm.
Pronation	Forearm rotates so palm faces downward/posteriorly in anatomical position.	Turning the palm down.

8.8 Planes and axes

Plane	Divides body into	Main movements / axis
Sagittal plane	Left and right portions.	Flexion/extension; movement around a transverse (frontal) axis.
Frontal / coronal plane	Front and back portions.	Abduction/adduction; movement around an anteroposterior axis.
Transverse / horizontal plane	Upper and lower portions.	Rotation; movement around a longitudinal/vertical axis.

Exam point: A plane is an imaginary flat surface in which movement occurs; an axis is an imaginary line around which rotation occurs.

CHAPTER 9 | Psychology and Sports

9.1 Meaning and importance

Sports psychology studies how thoughts, emotions and behaviour affect physical activity and performance, and how sport influences psychological wellbeing.

- Helps understand motivation, confidence, attention, arousal, anxiety, team relationships and responses to winning or losing.
- Supports goal setting, concentration, communication, coping strategies and consistent performance.

9.2 Personality

Personality is the relatively consistent pattern of thoughts, feelings and behaviours that makes an individual distinctive. It is influenced by biological, psychological and social factors.

Approach / theory	Core idea	Sports relevance
Trait approach	People show relatively stable characteristics or tendencies.	May help describe tendencies such as sociability or persistence, but does not perfectly predict behaviour.
Type approach	Groups people into broad categories.	Simple to understand but can oversimplify individual differences.
Psychodynamic approach	Behaviour can be influenced by conscious and unconscious processes and early experiences.	Highlights that behaviour may have deeper emotional influences.
Social-learning approach	Behaviour is learned through observation, reinforcement and social environment.	Coaching climate, role models and feedback can shape habits.
Interactional approach	Behaviour reflects interaction between personal traits and the situation.	An athlete may act differently under pressure, with different teammates or coaching styles.

9.3 Motivation

Motivation is the process that energises, directs and sustains behaviour toward a goal.

Type	Meaning	Example
Intrinsic	Driven by interest, enjoyment, mastery or personal satisfaction.	Practising a skill because improving it feels rewarding.
Extrinsic	Driven by external rewards, recognition, selection or consequences.	Training to earn a medal or team place.

Techniques to improve motivation

- Set SMART goals: specific, measurable, achievable, relevant and time-bound.
- Use positive, specific feedback focused on controllable actions and effort.
- Break a large goal into short-term process goals.
- Provide autonomy, choice, a supportive climate and appropriately challenging tasks.
- Use imagery, self-talk and progress records when helpful.

9.4 Aggression in sports

Aggression is behaviour intended to harm or intimidate another person. Anger is an emotion and does not always lead to aggressive behaviour.

Type	Description	Example
Hostile / reactive aggression	Harm is the main intention, often associated with anger.	Deliberately striking an opponent out of anger.
Instrumental aggression	Aggressive behaviour is used as a means to another goal, such as gaining advantage.	A deliberate rule-breaking act intended to stop an opponent; still unacceptable if against rules.

Assertiveness is firm, energetic, rule-compliant play without the intention to harm. Coaches can reduce harmful aggression through clear rules, emotional regulation, role modelling, fair officiating and consequences.

9.5 Exercise adherence

Exercise adherence means continuing to follow a physical-activity plan over time.

Factors that help adherence

- Enjoyment, achievable goals, social support, access to facilities, convenient timing and visible progress.
- A plan that matches ability, interests, health, culture and daily routine.
- Variety, feedback, self-monitoring and supportive coaching.

Barriers

- Lack of time, cost, transport, unsafe spaces, illness, low confidence, boredom, lack of support and unrealistic plans.

Strategies: schedule activity, start at a manageable level, choose enjoyable options, plan for barriers, use a partner/group if desired and review progress without shame.

9.6 Psychological attributes in sport

Attribute	Meaning / way to develop
Attention	Selecting relevant information and maintaining focus; use clear cues, routines and practice under realistic distractions.
Concentration	Sustained focus on the task; pre-performance routines and process goals can help.
Resilience	Adapting and recovering after setbacks; reflect on controllable factors, seek support and treat errors as feedback.
Mental toughness	Maintaining effort, composure and commitment under challenge; build through preparation, coping skills and realistic pressure practice—not by ignoring injury or distress.
Self-confidence	Belief in one's ability to perform a task; grows through preparation, mastery experiences, constructive feedback and realistic self-talk.
Arousal control	Managing activation level so it supports performance; slow breathing, relaxation, energising routines or imagery may help different athletes.

Exam point: Remember: motivation explains why effort begins and continues; adherence is maintaining activity over time; aggression intends harm, while assertiveness does not.

CHAPTER 10 | Training in Sports

10.1 Sports training

Sports training is a systematic, planned process of exercise and practice designed to improve performance, fitness, skills and readiness for competition.

10.2 Principles of sports training

Principle	Explanation
Specificity	Training should match the sport, movement, energy system and performance goal.
Overload	The body must receive a suitable challenge beyond its usual level to stimulate adaptation.
Progression	Increase training demands gradually as the athlete adapts.
Individualisation	Adjust training to the athlete's age, ability, health, training history and response.
Reversibility	Fitness gains can diminish when training is reduced or stopped for a period.
Variation	Vary activities and loads to maintain engagement and reduce monotony.
Rest and recovery	Adaptation requires sufficient recovery, sleep and nutrition; more training is not always better.
Continuity	Regular, consistent training supports improvement over time.
Specific adaptation	The body adapts to the type and dose of stress applied; the training stimulus should match the goal.

10.3 Training load, overload, adaptation and recovery

Training load is the total stress imposed by training. It can include volume (amount), intensity (difficulty), frequency (sessions) and density (work-to-rest relationship).

- **Overload:** a training stimulus greater than the body is accustomed to, but appropriate to the athlete's capacity.
- **Adaptation:** changes that occur as the body responds to repeated training, such as improved endurance or strength.
- **Recovery:** the period and processes through which the body restores and adapts after exertion.
- **Overtraining / excessive load:** prolonged imbalance between training stress and recovery can lead to persistent fatigue, declining performance, mood or sleep changes and increased injury/illness risk.

10.4 Warming up and limbering down

Type	Purpose and examples
General warm-up	Gradually raises body temperature and prepares the cardiovascular and muscular systems; e.g., easy jogging or dynamic whole-body movement.
Specific warm-up	Uses movements related to the sport or session; e.g., progressive passing drills before a ball game.
Dynamic mobility	Controlled movement through a comfortable range; commonly used before performance.
Cool-down / limbering down	Gradual reduction in activity after exercise; may include easy movement and gentle mobility/relaxation.
Static stretching	A position is held without bouncing; often used for flexibility work or after activity. It should not be painful.

Warm-up can improve readiness and allow gradual increases in intensity. Cool-down helps the transition to rest, although it does not guarantee prevention of all soreness or injury.

10.5 Skill, technique, tactics and strategy

Term	Meaning	Example
Skill	Ability to perform a task effectively, developed through learning and practice.	Dribbling a basketball.
Technique	The specific method or movement pattern used to perform a skill.	Footwork and release mechanics in a basketball shot.
Tactics	Immediate decisions or actions used to gain an advantage in a particular situation.	Changing direction to create space against a defender.
Strategy	The broader plan for achieving a competitive objective.	A team plan to press high or conserve energy until late in a match.

10.6 Talent identification and talent development

Talent identification is the process of recognising individuals who show characteristics suggesting potential in a sport. **Talent development** is the longer-term process of providing coaching, practice, competition, support and opportunities so potential can grow.

- Consider physical, technical, tactical, psychological and learning characteristics—not a single test or early result.
- Growth and maturation differ between young people; early physical advantage does not guarantee long-term success.
- Fair, inclusive selection considers access to coaching, motivation, health, opportunity and development over time.

10.7 Training methods for fitness components

A. Strength

Strength is the ability of muscles to exert force against resistance.

- **Isometric:** muscle produces force without visible joint movement, e.g., a wall sit or plank.
- **Isotonic:** muscle changes length while moving a load; includes concentric (shortening) and eccentric (lengthening) actions.
- **Isokinetic:** movement occurs at a controlled constant speed using specialised equipment.
- Resistance training should use correct technique, suitable loads, gradual progression and supervision appropriate to age and experience.

B. Endurance

Endurance is the ability to sustain activity and resist fatigue.

- **Continuous training:** steady activity without planned rest; useful for aerobic base development.
- **Interval training:** work periods alternate with planned recovery; intensity and recovery are adjusted to the goal.
- **Fartlek training:** continuous activity with changes in pace or terrain ('speed play').
- **Circuit training:** a sequence of exercise stations targeting different components; rest and station duration can be varied.

C. Speed

Speed is the ability to perform a movement or cover a distance in minimal time.

- Develop through appropriate sprint technique, reaction practice, short high-quality repetitions and sufficient recovery.
- Speed training should stop or reduce when technique deteriorates markedly; fatigue-heavy repetitions may train a different quality.

D. Flexibility

Flexibility is the range of motion available at a joint or series of joints.

- **Static stretching:** hold a comfortable position; avoid bouncing and sharp pain.
- **Dynamic stretching:** controlled movement through range; often suited to warm-up.

- **PNF:** proprioceptive neuromuscular facilitation, a method combining muscle contraction and stretching, generally best learned with qualified guidance.

E. Coordinative abilities

Coordination is the ability to organise movements accurately and efficiently. Common coordinative abilities include balance, reaction, orientation, rhythm, differentiation, coupling and adaptation.

- Practice tasks that challenge timing, rhythm, balance, reaction and movement adjustment in a safe, progressive way.

10.8 Circuit training

Circuit training involves completing a series of exercise stations in a planned order. Stations may target strength, endurance, speed, mobility or coordination.

- **Planning:** identify the objective; choose exercises; decide station order, work time/repetitions, rest, rounds and equipment; demonstrate safe technique.
- **Advantages:** efficient use of time and space, variety, adaptable difficulty, multiple fitness components and group participation.
- **Limitations:** technique may deteriorate when rushed; equipment or supervision may be limited; not every circuit suits every athlete or goal.

10.9 Training plan: simple structure

Stage	What to include
Goal	Specific performance or fitness objective.
Baseline	Current ability and relevant health/safety considerations.
Plan	Exercise selection, frequency, intensity, duration/volume and progression.
Recovery	Rest days, sleep, nutrition and lighter sessions.
Monitoring	Record effort, performance, fatigue, mood and symptoms.
Review	Adjust the plan based on response and goal progress.

Exam point: Training is not simply 'working harder'. Effective training matches the goal, increases demands gradually and allows recovery.

FINAL REVISION SECTION

High-yield formulas and facts

Topic	Formula / fact
Knock-out matches	N – 1
Byes in knock-out	Next power of 2 – N (if N is not already a power of 2)
Single league matches	N(N – 1) ÷ 2
Double league matches	N(N – 1)
BMI	Weight (kg) ÷ [height (m)]^2
Cardiac output	Heart rate × stroke volume
Newton's second law	Force = mass × acceleration
Planes	Sagittal: flexion/extension; frontal: abduction/adduction; transverse: rotation
Lever classes	1st: fulcrum in middle; 2nd: resistance in middle; 3rd: effort in middle
Sprain vs strain	Sprain = ligament; strain = muscle/tendon
Female Athlete Triad	Low energy availability/disordered eating, menstrual dysfunction, low bone mineral density
Test vs measurement vs evaluation	Procedure/task; numerical result; interpretation/judgement

Common compare-and-contrast questions

Pair	Difference
Intramural vs extramural	Within one institution vs between institutions.
Knock-out vs league	One loss eliminates vs each team plays all/most opponents according to format.
Test vs measurement	Test is the task; measurement is the score obtained.
Flexion vs extension	Joint angle decreases vs increases/straightens.
Kinematics vs kinetics	Describes motion vs studies forces causing or affecting motion.
Intrinsic vs extrinsic motivation	Internal satisfaction vs external reward/consequence.
Skill vs technique	Ability to perform a task vs the method used to perform it.
Tactics vs strategy	Situation-specific action vs broader plan.
Sprain vs strain	Ligament injury vs muscle/tendon injury.

Answer-writing template

- **1 mark:** give a precise definition or one direct fact.
- **2–3 marks:** definition + two/three explained points or a definition + example.
- **4–5 marks:** brief introduction, organised headings/numbered points, explanation, relevant example/table/diagram, and a concluding significance statement.
- For a numerical question: write the formula, substitute values with units, show calculation and state the final answer.
- For a diagram: use a clear title, straight labels, correct directional arrows and a brief explanation.

Glossary

Term	Meaning
Adaptation	Change in response to repeated training or environmental demands.
Aerobic	Energy production involving oxygen-dependent processes.
Anaerobic	Energy production that does not rely directly on oxygen for the immediate process.
Arousal	Level of physiological and psychological activation.
Biomechanics	Application of mechanical principles to living bodies and movement.
Fixture	Schedule/pairing arrangement for tournament matches.
Hypertrophy	Increase in muscle size.
Inclusion	Meaningful participation by people with different abilities, supported by accessible environments and adaptations.
Overload	A suitable increase in training demand above the accustomed level.
Recovery	Restoration and adaptation following activity.
Seeding	Placing strong/ranked teams in separate bracket positions to reduce early meetings.
Somatotype	Classification of body build; traditional categories include endomorph, mesomorph and ectomorph.
Wellness	Multidimensional state of wellbeing supported by healthy habits and social/environmental factors.

Syllabus and source note

Syllabus basis: CBSE, Physical Education (Subject Code 048), Class XI–XII, Curriculum 2026–27, official curriculum PDF: CBSE official syllabus PDF. The unit headings and required topic coverage in these notes follow the Class XII syllabus checklist. Explanations and examples are original study aids; the school-prescribed textbook, teacher's directions, current practical manuals and latest exam circulars should be used for exact protocols and scoring.

Health and safety note: These notes are for education, not individual medical advice. Exercise, yoga, injury care, nutrition and posture management should be adapted to the individual. Seek a qualified professional for pain, illness, suspected injury or personal health concerns.